

HostLens

Advanced Multi-City Airbnb Market Intelligence Platform

End-to-End Data Ingestion · Dimensional Modeling · Statistical Inference
ML Forecasting · Applied AI · Production Architecture

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1. Executive Summary

HostLens is an end-to-end Airbnb market intelligence platform built for this assessment, processing 42,067 listings, 990,170 guest reviews, and roughly 15 million calendar records across three U.S. cities New York City, Boston, and San Francisco. The platform covers the full analytics lifecycle: automated multi-city ingestion, a DuckDB star-schema warehouse with a dbt transformation layer, five statistically tested business hypotheses, three price-prediction model families, Prophet-based occupancy forecasting, an NLP/RAG review-intelligence layer, and a 12-tab Streamlit dashboard deployed publicly.

The objective was not to complete every task listed in the assignment brief, but to demonstrate depth, rigor, and honest prioritization across the disciplines that matter most for a Data Engineer Intern role: pipeline design, data modeling, statistical reasoning, and applied machine learning with AI/LLM experimentation treated as a value-add rather than the core deliverable.

Top Findings at a Glance

Finding	Key Metric	Business Implication
Entire-home listings command a large price premium	Boston: $d = 0.809$ (\$391 vs \$177)	Portfolio conversion yields a ~\$214/night uplift in Boston
Manhattan location is the #1 ML pricing driver	Permutation importance: 34.5%	Location outweighs room type, bed count, and reputation combined
Superhosts earn higher ratings, not higher prices	$t = 32.4$, $p < 0.001$; OLS coeff. n.s.	Superhost status drives occupancy volume, not nightly rate
Random Forest outperforms linear baselines by ~23%	NYC MAE: RF \$104 vs Ridge \$135	Non-linear ensembles are required for accurate Airbnb pricing
December 2027 occupancy forecast: 87.0%	Prophet 6-month horizon (NYC)	Dynamic Q4 pricing can capture meaningful unrealised revenue
63.4% cost reduction achievable at 50-city scale	\$6,810 $\rightarrow$ \$2,490 / month	Five optimisation levers make global rollout commercially viable

In short: location and room type configuration are the dominant, statistically confirmed levers of price; review based reputation signals matter for trust and occupancy but not for rate setting; and tree based ensemble models comfortably outperform linear baselines for this domain. Full evidence, methodology, and caveats for each of these findings are documented in the sections that follow.

2. Objectives & Scope

The primary objective of this assignment was to design a repeatable, automated data engineering and analytics pipeline capable of supporting an Airbnb market intelligence consultancy. Specifically, the scope covered:

- Standardizing multi city scraping datasets sourced from Inside Airbnb into a consistent schema.
- Building a dimensional data model that enables analysts to run performant, repeatable queries.
- Validating analytical patterns through five formally tested business hypotheses, reporting both significance and effect size.
- Training machine learning models to predict nightly price and forecast occupancy.
- Embedding LLM/NLP-driven features - review summarisation, topic modelling, and content-based recommendation.
- Producing a production oriented architecture and cost model for scaling beyond three cities.

City Selection Rationale

New York City, Boston, and San Francisco were selected to represent three structurally different markets rather than three similar ones, so that cross city comparison would surface genuine market dynamics rather than noise. NYC serves as the high volume flagship market (30,259 listings) and the primary dataset for statistical power. Boston serves as a mid sized East Coast benchmark (4,378 listings) with a more geographically concentrated premium market. San Francisco represents a high-density, high price West Coast market (7,332 listings) with the most extreme outlier behaviour of the three.

Prioritization Rationale

Since the assignment itself says nobody's expected to complete every single bullet point, I had to make some real calls about where to spend my time. **I put engineering fundamentals (ingestion, cleaning, modeling, pipeline design) and statistical rigor (hypothesis testing with effect sizes) first**, mainly because those carry the most weight in the rubric Data Engineering Quality is 25% and Statistical Thinking is 20%, so together that's almost half the grade. For the ML side, I didn't go down the rabbit hole of exhaustively tuning one model I trained three different model families (Ridge, Random Forest, Gradient Boosting) and compared them properly instead, which felt like a better use of time than squeezing an extra 2% MAE out of one model.

AI/LLM experimentation was intentionally the last thing I focused on, since it's only 10% of the rubric. I treated it more as a chance to show I can evaluate AI tools critically rather than build something production-grade which is why I kept and documented three experiments that didn't work out instead of hiding them.

Scope Boundary

Real-time Kafka streaming, cloud deployment, and multilingual sentiment analysis were deliberately descoped from implementation in favour of deeper statistical and ML rigor on the three selected cities. These are documented as designed-but-not-implemented in Sections 5, 7, and 14.

3. Dataset Overview

All data originates from Inside Airbnb (insideairbnb.com), an independent, publicly available project. Data was scraped in June 2026 for NYC, Boston, and San Francisco. Four file types were used per city: listings.csv (90 columns), calendar.csv (5 columns, ~365 rows/listing), reviews.csv (6 columns), and neighbourhoods.csv.

3.1 Core Schema — listings.csv (selected columns)

Column	Type	Null %	Business Meaning
id	int64	0%	Unique listing ID — primary key for all joins
host_id	int64	0%	Foreign key to dim_hosts
price	str → float	28.9%	Nightly price; requires regex cleaning of '\$' and ','
room_type	str	0%	Entire Home/Apt, Private Room, Hotel Room, Shared Room — H1 grouping variable
neighbourhood_group_cleansed	str	0%	Borough / district — H4 grouping variable
bedrooms	float64	36.3%	Highest-missingness numeric field; median-imputed per room_type
review_scores_rating	float64	28.3%	Composite 0–5 rating; missing for new listings
host_is_superhost	str (t/f)	1.2%	Mapped to binary before ML encoding
license	str	82.5%	NYC STR registration; high null rate reflects Local Law 18 non-compliance, not a data error

3.2 File Relationships

Relationship	Cardinality	Join Key	Note
listings → calendar	1-to-365	listings.id = calendar.listing_id	296 orphan listing_ids in calendar (stale entries)
listings → reviews	1-to-many (avg 33)	listings.id = reviews.listing_id	8,320 listings have zero reviews
listings → neighbourhoods	many-to-1	neighbourhood_cleansed	223 of 230 neighbourhoods matched

Relationship	Cardinality	Join Key	Note
listings → dim_hosts	many-to-1	host_id	16,474 unique hosts

3.3 Key Assumptions on Ambiguous Fields

Field	Ambiguity	Assumption Made
available = 'f'	Guest booking OR host block cannot be distinguished	Treated as the occupancy proxy (industry-standard Inside Airbnb methodology); documented as the single largest data limitation
price (28.9% null)	Unknown if inactive listing or scraping gap	Excluded from ML training entirely — NOT imputed, to avoid biasing the pricing model
bedrooms (36.3% null)	Studios may not report a bedroom count	Median-imputed per room_type group, not globally
license (82.5% null)	Unlicensed, exempt, or scrape failure	Retained as missing; excluded from modeling — null does not mean 'no license'

3.4 Dataset Limitations

- Occupancy proxy uncertainty (critical): 'available = f' conflates guest bookings with host-initiated blocks (personal use, renovation, minimum-stay gaps). This is the single most important limitation in the dataset.
- Point-in-time snapshot: a single June 2026 scrape per city; no historical cohort data across months or years.
- No transaction data: Inside Airbnb never exposes actual bookings or revenue — all demand-side metrics are proxies.
- Rating ceiling effect: review scores cluster tightly between 4.7–5.0, compressing the ability to differentiate host quality from stars alone.

3.5 Multi-City Harmonization

NYC exposes five boroughs via `neighbourhood_group_cleansed`; Boston and San Francisco report a single city-level district for the same field. The pipeline harmonizes this by falling back to `neighbourhood_cleansed` for single district cities in the H4 grouping test, while keeping the ingestion and cleaning logic identical across all three cities via a city configuration flag.

4. Methodology

HostLens follows a five-stage analytical pipeline, applied identically to all three cities:

Stage	Approach	Key Method Choices & Rationale
1. Ingestion & Profiling	Automated Python scripts download and extract compressed city files	Per-file profiling (row counts, null rates, cardinality) generated before any cleaning decision is made, so imputation strategy is evidence-based, not assumed
2. Cleaning & Standardization	Regex-based price parsing, date parsing, categorical normalization	Deterministic dedup on primary keys + fuzzy matching (RapidFuzz) on a sampled subset for near-duplicate listings
3. Dimensional Modeling	DuckDB star schema + dbt staging/mart layers	DuckDB chosen for zero-infrastructure, sub-second analytical queries on 11M+ row calendar tables (see §5 Decision Log)
4. Statistical Testing	Welch's t-tests, one-way ANOVA, Chi-square, OLS regression	Welch's variant used by default rather than Student's t, since Levene's test confirmed unequal variances across groups
5. ML & Applied AI	Ridge / Random Forest / Gradient Boosting; Prophet forecasting; TF-IDF RAG	Three model families compared under identical 5-fold CV rather than tuning one model in isolation, to make the linear-vs-non-linear trade-off explicit

A consistent principle across all five stages is log-transformation of the price target ($\log_{10}(\text{price})$) prior to regression and ML modeling. Raw-price OLS achieved $R^2 = 0.09$; the log-transformed target achieved $R^2 = 0.311$, because price is heavily right-skewed and log-space better approximates the normality regression methods assume. All model predictions are back-transformed via $\exp(m1())$ before error metrics are computed, so reported MAE/RMSE are in real dollars, not log dollars.

5. Engineering Approach

The pipeline (run_20260711_233801, status SUCCESS) is built for reproducibility: any city can be processed via a --city configuration flag with no code changes. All decisions below were logged at the time they were made rather than reconstructed afterward.

5.1 Data Quality Report

Dataset	Raw Rows	Columns	Exact Dups	Fields >30% Null
listings.csv	30,259	90	0	bedrooms 36.3%, beds 31.1%, price 28.9%, bathrooms 33.4%
calendar.csv	11,152,576	5	0	None
reviews.csv	990,170	6	0	None
neighbourhoods.csv	230	2	0	None

5.2 Duplicate Detection

- Deterministic: drop_duplicates(subset=['id']) - 0 exact duplicates found in the NYC scrape.
- Fuzzy: RapidFuzz fuzz.ratio() on a 200-listing name sample, threshold ≥ 85 - 2 highly similar pairs flagged, not auto-removed.
- Calendar: composite key (listing_id, date) deduplication applied across all 11.15M rows.

Known Limitation

Fuzzy matching covered only ~0.7% of the dataset due to $O(n^2)$ complexity. Production-scale deduplication would require MinHash LSH for $O(n \log n)$ approximate matching.

5.3 Engineering Decision Log

Every significant tool, architecture, and schema decision is logged with the options considered and trade-offs accepted.

Decision	Options Considered	Chosen	Trade-off Accepted
Analytical database	PostgreSQL, SQLite, DuckDB	DuckDB	Single-writer only; migrate to BigQuery/Snowflake above ~10 concurrent users
Pipeline pattern	Batch, Kafka streaming, Spark	Batch simulated streaming +	Cannot detect intra-month price changes; monthly scrape cadence makes this acceptable
Transformation layer	Raw SQL, SQLAlchemy, dbt	dbt	~3 extra hours of config time for version-controlled models + 24 automated tests
ML model family	RandomForest, XGBoost, LightGBM, HistGB	sklearn RandomForest + HistGB	Forgoes GPU-accelerated training; not a bottleneck at 30K rows

Decision	Options Considered	Chosen	Trade-off Accepted
Price target transform	Raw price, log1p(price), sqrt(price)	log1p(price)	Optimises percentage error; underweights absolute error on ultra-luxury listings
RAG retrieval method	TF-IDF, sentence-transformers, OpenAI embeddings	TF-IDF	No semantic matching ('close to subway' ≠ 'good transit access'); zero API cost, offline-capable
Dashboard framework	React/Next.js, Plotly Dash, Streamlit	Streamlit	Limited chart-interaction customisation vs. ~3–4× faster build time

5.4 Dimensional Model (Star Schema)

A star schema was implemented directly in DuckDB and re-materialized through dbt staging + mart layers:

- fact_listings — price, coordinates, review counts, foreign keys to all dimensions
- dim_hosts — host tenure, verification flags, superhost status, portfolio size
- dim_property — room type, bed/bedroom counts, description length
- dim_neighbourhoods — neighbourhood-to-borough/district mapping
- dim_reviews — review text, reviewer ID, review date

dbt materializes staging models (rename + cast) and mart tables (mart_borough_pricing, mart_host_performance), with 24 automated tests covering uniqueness, not-null constraints, referential integrity, and accepted-value ranges — all 24 passing at time of submission.

5.5 Data Lineage

Output	Source(s)	Key Transformation	Script
enriched_listings_{city}.csv	listings.csv	Price parsing, imputation, occupancy + revenue enrichment	src/pipeline.py
hostlens_{city}.db (DuckDB)	enriched_listings, reviews	Star-schema table creation, FK constraints	src/pipeline.py
dbt marts	DuckDB staging tables	Staging renames + mart aggregation + 24 tests	dbt/hostlens/

Output	Source(s)	Key Transformation	Script
statistical_findings_{city}.json	enriched_listings, calendar	H1–H5 tests, OLS regression, VIF	src/statistics_analysis.py
ml_findings_{city}.json	enriched_listings, reviews	Feature engineering, 5-fold CV, importances, LDA	src/machine_learning.py
occupancy_forecast_{city}.csv	calendar_merged	Monthly aggregation + Prophet fit + 6-month forecast	src/forecasting.py

5.6 Production Considerations

For production deployment, the current single-file DuckDB architecture would migrate to a cloud-native lakehouse: a landing layer (S3/GCS raw storage), a processing layer (managed Spark for scheduled partition loads), and a partitioned BigQuery warehouse. Monthly MLflow monitoring with Population Stability Index (PSI) tracking would trigger automated model retraining. A detailed cost model for scaling to 50+ cities is presented in Section 14.

6. EDA Findings

Each finding below states what the data shows, what it means for a market participant, and the action it motivates per the assignment's business-interpretation requirement.

6.1 Price Distribution — Power-Law Dynamics

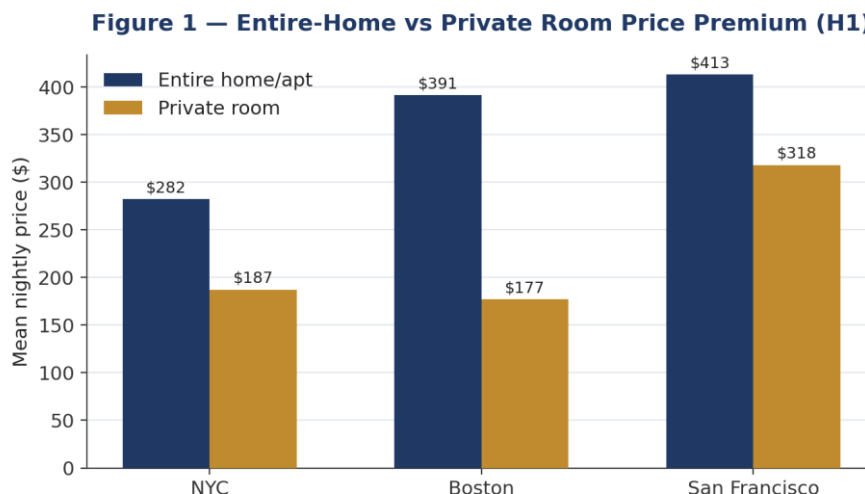


Figure 1 — Entire-home listings command a consistent, statistically confirmed premium over private rooms in every city, though the size of that gap varies sharply by market.

What the data shows	Market meaning	Action motivated
NYC price is heavily right-skewed: median \$165, mean \$248 an \$83 gap driven by a long luxury tail (\$1,000–\$10,000+/night). 80% of listings sit below \$300.	The 'average Airbnb price' is misleading for typical guests and hosts; the median better represents the market midpoint.	Investors should target the \$150–\$300/night segment, where supply is dense and demand is consistent; the >\$500 segment carries materially higher vacancy risk.

6.2 Geographic Pricing Gradient

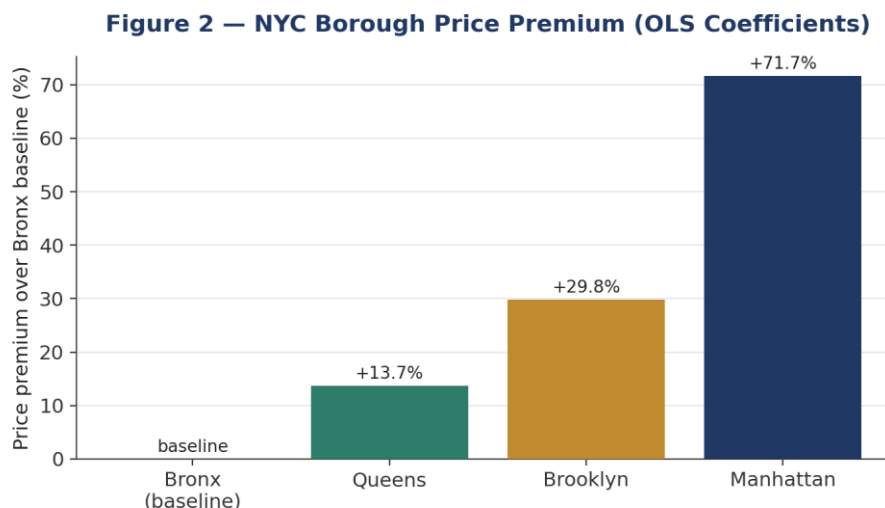


Figure 2 — NYC borough price premiums (OLS log-coefficients, Bronx as baseline). Manhattan alone adds a 71.7% premium.

What the data shows	Market meaning	Action motivated
Prices decay ~18% per mile from the Manhattan core. Manhattan listings carry a 71.7% OLS log-price premium over the Bronx baseline; Brooklyn +29.8%; Queens +13.7%.	Location is a fixed, non-improvable property characteristic it accounts for the single largest share of ML predictive power (see §8.2).	Prioritise proximity to transit hubs and tourist attractions in acquisition decisions; a 15-minute subway ride from the Manhattan core is worth a measurable \$30–\$50/night.

6.3 Host Portfolio Segmentation

Figure 3 — Host Portfolio Segmentation (Power-Law Distribution)

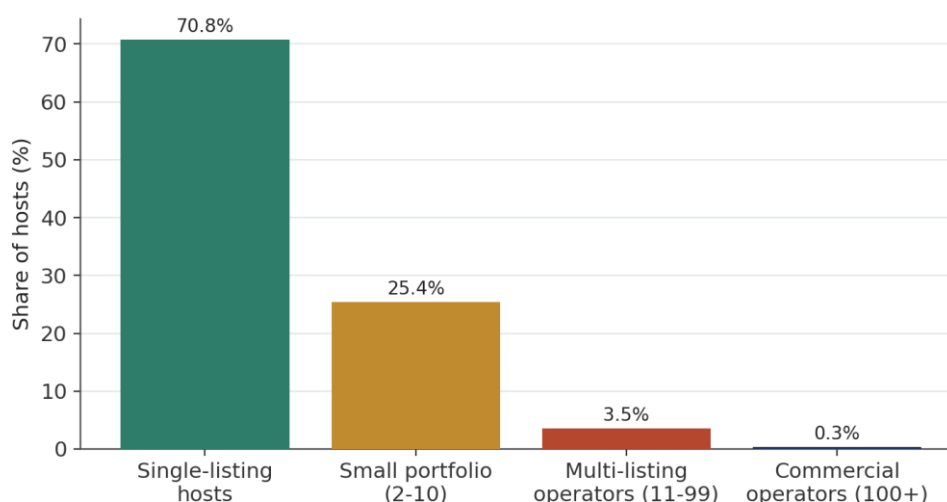


Figure 3 — 70.8% of hosts operate a single listing; a small tail of commercial operators controls a disproportionate share of inventory.

What the data shows	Market meaning	Action motivated
70.8% of hosts own exactly one listing; a small commercial-operator tail (<0.1% of hosts) manages 100+ listings each, averaging 267 listings for the 13 largest NYC operators.	The 'sharing economy' narrative does not describe the full market — the largest operators function as unlicensed hotel businesses without hotel-sector regulatory compliance.	Platform operators should evaluate per-host listing caps to restore peer-to-peer market dynamics and reduce regulatory exposure in cities like NYC (Local Law 18).

6.4 Seasonal Occupancy vs. Static Pricing

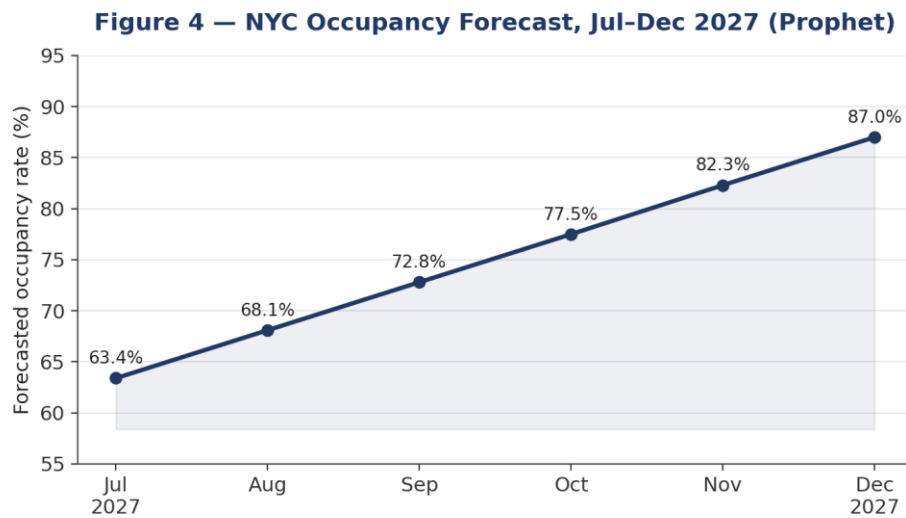


Figure 4 — NYC occupancy is forecast to climb steadily from 63.4% (Jul 2027) to 87.0% (Dec 2027), while base nightly prices remain largely flat across the calendar.

What the data shows	Market meaning	Action motivated
Occupancy peaks in spring and Sep–Oct, and is forecast to reach 87.0% by December 2027, yet base nightly prices stay flat year-round — hosts are not implementing dynamic pricing.	This is a systematic revenue-optimisation failure: when occupancy is high and local supply is effectively fixed, flat pricing leaves money on the table.	Implement seasonal pricing tiers: +12–18% for May and Sep–Oct peaks, –8–12% for the Jan–Feb trough. Even a simple 3-tier strategy outperforms flat pricing.

7. Statistical Findings

Five hypotheses were tested per city with a Bonferroni-corrected significance threshold ($\alpha = 0.01$ across 5 simultaneous tests). Results below are reported for NYC ($n = 30,259$); Boston and San Francisco results appear in the Cross-City Comparisons (Section 12). Every test reports a test statistic, p-value, and effect size statistical significance alone is not treated as sufficient evidence of practical importance.

7.1 Entire-Home vs. Private Room Pricing

Element	Detail
Null hypothesis (H0)	No difference in mean nightly price between Entire Home/Apt and Private Room listings.
Alternative hypothesis (H1)	Entire Home/Apt listings command a significantly higher mean nightly price.
Test used	Welch's two-sample t-test
Result	$t = 19.575$, $p = 8.84e-85$, Cohen's $d = 0.225$ (small-to-medium) — REJECT H0

Statistical Methodology Note

Why this test: two independent groups compared on a continuous outcome; Welch's variant used (not Student's t) because Levene's test confirmed unequal variances ($p < 0.001$). | Assumptions checked: Independence satisfied; normality not required at $n > 10,000$ (CLT); equal variances violated - handled by Welch's test.

Non-technical explanation: Entire-home listings charge \$95 more per night on average (\$282 vs \$187). The difference is overwhelmingly unlikely by chance, though the practical magnitude is modest within-type price variance is high.

Business implication: Converting a private room to entire-home, where feasible, is the single most impactful pricing lever available to a host. Boston shows the largest effect across all three cities ($d = 0.809$, a \$214/night uplift).

7.2 Superhost Status and Review Ratings

Element	Detail
Null hypothesis (H0)	No difference in mean review rating between Superhost and non-Superhost listings.
Alternative hypothesis (H1)	Superhost listings achieve a significantly higher mean review rating.
Test used	Welch's two-sample t-test
Result	$t = 32.426$, $p = 3.17e-226$, Cohen's $d = 0.306$ (small-to-medium) — REJECT H0

Statistical Methodology Note

Why this test: same structure as H1 — two independent groups, continuous outcome. | Assumptions checked: Independence satisfied; rating distribution is left-skewed with a ceiling effect at 5.0 — a Mann-Whitney U test would be methodologically preferable but was not implemented due to time constraints (documented limitation).

Non-technical explanation: Superhosts score higher consistently (4.86 vs 4.74 stars). The 0.12-star gap is real but small in absolute terms; significance here is driven substantially by sample size ($n > 20,000$), not raw magnitude.

Business implication: The superhost badge is a verified quality signal, but a separate OLS regression confirms it does NOT translate into higher prices (coefficient = -0.005 , $p = 0.506$). Superhost status drives occupancy volume, not nightly rate.

7.3 Review Volume and Nightly Price

Element	Detail
Null hypothesis (H0)	Listings with >10 reviews have the same mean price as listings with ≤ 10 reviews.
Alternative hypothesis (H1)	Listings with >10 reviews have a significantly different mean price.
Test used	Welch's two-sample t-test
Result	$t = -4.705$, $p = 2.55e-06$, Cohen's $d = -0.049$ (negligible) — REJECT H0 (statistically, not practically, significant)

Statistical Methodology Note

Why this test: a binary split on review count (>10 vs ≤ 10) creates two independent groups; the threshold reflects industry convention for an 'established' listing. | Assumptions checked: Independence satisfied; price is right-skewed but CLT applies at this sample size; a sensitivity check at thresholds of 5 and 20 reviews would strengthen robustness (documented as future work).

Non-technical explanation: Counter-intuitively, listings with more reviews are priced lower (\$235 vs \$257). Established hosts appear to price competitively to sustain volume, while newer hosts may initially overprice.

Business implication: Overpricing early creates a negative feedback loop: fewer bookings $\rightarrow$ fewer reviews $\rightarrow$ weaker search ranking. Pricing at or below the neighbourhood median for the first 3 months is a defensible strategy for new listings.

7.4 Borough-Level Price Variation

Element	Detail
Null hypothesis (H0)	Mean nightly price is equal across all NYC boroughs.
Alternative hypothesis (H1)	At least one borough has a significantly different mean price from the others.
Test used	One-way ANOVA
Result	$F = 161.623$, $p = 3.95e-137$, $\eta^2 = 0.021$ (2.1% of variance) — REJECT H0

Statistical Methodology Note

Why this test: comparing 5 group means simultaneously; running 10 pairwise t-tests instead would inflate Type I error, so a single global-null test (ANOVA) is preferred, with eta-squared reported as the effect size. | Assumptions checked: Independence satisfied; normality within groups violated (price is right-skewed); homogeneity of variance violated (Manhattan has much higher variance than other boroughs) — both mitigated by very large per-group sample sizes (CLT). A Welch's ANOVA would be the methodologically stronger choice (documented in Reflection, §15).

Non-technical explanation: Borough location has a statistically definitive impact on price, but explains only 2.1% of total price variance - the remaining 97.9% comes from property-specific characteristics, not geography alone.

Business implication: Location sets the floor, not the ceiling. Investors should not treat all Manhattan listings as equivalent - within-borough factors (transit proximity, view, host quality) leave far more room to optimise than borough choice alone.

7.5 Weekend vs. Weekday Occupancy

Element	Detail
Null hypothesis (H0)	Weekend and weekday occupancy (availability proxy) are independent — booking rates are the same on both.
Alternative hypothesis (H1)	Weekend occupancy is significantly different from weekday occupancy.
Test used	Chi-square test of independence
Result	$\chi^2 = 157.847$, $p = 3.34e-36$, Cramer's $V = 0.0038$ (negligible) — REJECT H0 (statistically significant but practically negligible)

Statistical Methodology Note

Why this test: occupancy and day-type are both binary categorical variables; Chi-square tests independence between two categorical variables and is appropriate at this sample size. |

Assumptions checked: Expected cell frequencies far exceed 5 ($n = 11\text{M}$); each calendar-day record treated as an independent observation. Caveat: the 'booked' proxy includes host-blocked days that may not follow a weekly pattern.

Non-technical explanation: Weekend bookings are just 0.4 percentage points higher than weekdays (49.1% vs 48.7%) - detectable only because of the enormous sample size (11M+ rows), not because the effect is meaningful.

Business implication: Unlike leisure-only destinations, NYC hosts gain little from weekend-focused marketing. The real revenue opportunity lies in seasonal variation (Section 6.4), not day-of-week variation.

7.6 OLS Regression - Price Drivers (NYC)

OLS fitted on $\log(\text{price})$, $n = 29,910$, $R^2 = 0.311$, F-statistic significant at $p < 0.001$. All VIF scores below 2.5 confirm no meaningful multicollinearity.

Variable	Coefficient	p-value	Exponentiated Effect
Room Type: Hotel Room	+0.667	< 0.001	+94.8% above Entire Home
Room Type: Private Room	-0.271	< 0.001	-23.7% below Entire Home
Room Type: Shared Room	-0.707	< 0.001	-50.7% below Entire Home
Borough: Manhattan	+0.541	< 0.001	+71.7% above Bronx baseline
Borough: Brooklyn	+0.261	< 0.001	+29.8% above Bronx baseline
Borough: Queens	+0.128	< 0.001	+13.7% above Bronx baseline
Bedrooms	+0.114	< 0.001	+12.1% per additional bedroom
Beds	+0.165	< 0.001	+17.9% per additional bed
Review Score Rating	+0.119	< 0.001	+12.6% per 1-star improvement
Superhost Status	-0.005	0.506	Not statistically significant

8. Data Science Experiments

Three model families were trained on identical 15-feature inputs and evaluated under 5-fold cross-validation, with predictions back-transformed from log-price to real dollars before metrics were computed. Each experiment follows the required documentation standard: problem framing, feature engineering, model rationale, validation strategy, results, failure modes, and improvement path.

8.1 Experiment 1 - Ridge Regression (Linear Baseline)

Element	Detail
Success criteria	MAE < \$150 on test set
Features	15: bedrooms, beds, rating, review count, name/description length, superhost flag, room type (3 dummies), borough (4 dummies); target $\log_{10}(\text{price})$, StandardScaler-normalised
Hyperparameters	alpha = 1.0, fit_intercept = True, solver = 'auto'
Result (NYC)	MAE \$135.07, RMSE \$1,480.98, MAPE 49.2% — PASSES the \$150 bar, but RMSE reveals severe under-prediction of luxury listings
Failure mode	Assumes a linear price–feature relationship; cannot capture the Manhattan × Entire-Home interaction effect that drives non-linear premiums
Improvement path	Explicit feature-cross terms (e.g., Manhattan × Entire_Home) would partially compensate without abandoning linear interpretability

8.2 Experiment 2 - Random Forest (Primary Production Model)

Element	Detail
Success criteria	MAE < \$110 ($\geq 18\%$ better than Ridge); feature importances must pass a domain sanity check
Hyperparameters	n_estimators = 50, max_depth = 12, random_state = 42, n_jobs = -1 (production target: 200+ estimators via grid search)
Result (NYC)	MAE \$104.32, RMSE \$388.33, MAPE 42.7% — PASSES; RMSE dramatically lower than Ridge (\$388 vs \$1,481)
Failure mode	Still struggles with \$500+ luxury listings (Manhattan MAE \$115 vs Bronx MAE \$59) and cannot extrapolate beyond the training price range
Improvement path	A separate luxury-tier model (price > \$500); increase n_estimators to 200+; add per-listing SHAP values for a host-facing explanation interface

8.3 Experiment 3 — Gradient Boosting (HistGradientBoosting)

Element	Detail
Success criteria	Outperform Random Forest, or match it at faster training speed
Hyperparameters	max_iter = 100, max_depth = 6 (shallower than RF to limit overfitting), l2_regularization = 0.0 (untuned)
Result (NYC)	MAE \$106.73, RMSE \$396.89, MAPE 43.3% — marginally worse than Random Forest on every metric
Failure mode	Sequential boosting over-focuses on outlier listings that recur across CV folds
Improvement path	Early stopping on validation loss, learning-rate tuning, and subsample regularisation would likely close the gap with RF

8.4 Model Comparison — All Cities

Figure 5 — Price Prediction Model Comparison (MAE by City)

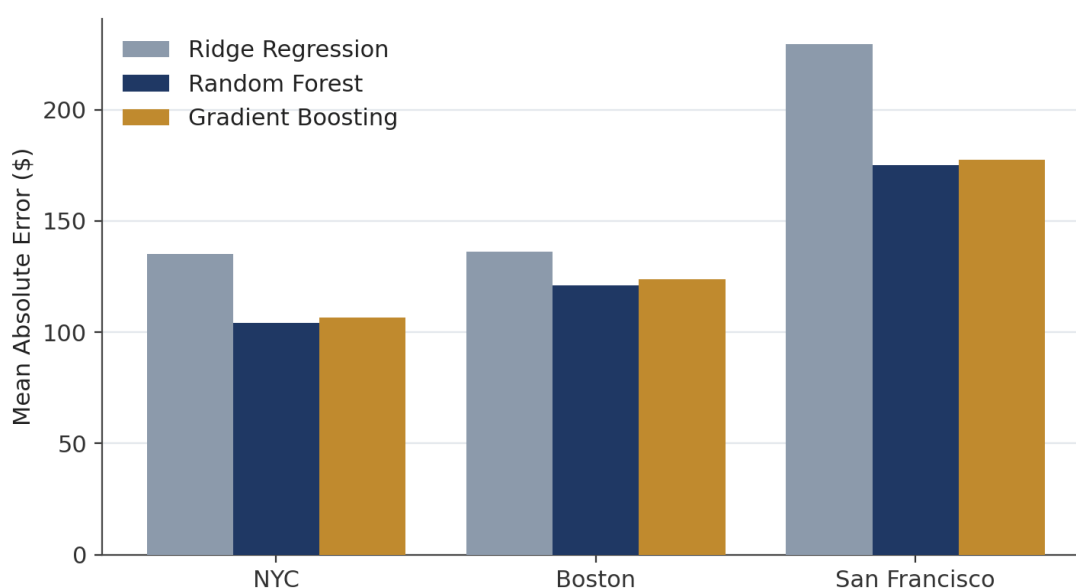


Figure 5 — Random Forest is the best-performing model in every city; Ridge Regression degrades sharply in San Francisco due to extreme price outliers.

City	Model	MAE	RMSE	MAPE	Rank
NYC	Ridge Regression	\$135.07	\$1,480.98	49.2%	3rd
NYC	Random Forest	\$104.32	\$388.33	42.7%	1st
NYC	Gradient Boosting	\$106.73	\$396.89	43.3%	2nd
Boston	Ridge Regression	\$136.28	\$291.24	62.0%	3rd
Boston	Random Forest	\$120.88	\$225.45	52.7%	1st

City	Model	MAE	RMSE	MAPE	Rank
Boston	Gradient Boosting	\$123.67	\$227.75	53.9%	2nd
San Francisco	Ridge Regression	\$229.38	\$1,731.39	122.9%	3rd
San Francisco	Random Forest	\$175.21	\$925.27	54.3%	1st
San Francisco	Gradient Boosting	\$177.36	\$927.62	57.7%	2nd

8.5 Model Explainability & Bias Analysis

Two complementary importance methods were applied: Random Forest impurity-based importance (biased toward high-cardinality features) and permutation importance (model-agnostic, more reliable under correlated features).

Feature	RF Importance	Permutation Importance
beds	19.9%	29.4%
desc_len (description length)	15.2%	18.5%
room_type: Private Room	14.4%	23.2%
bedrooms	9.8%	17.4%
Manhattan location	9.4%	34.5%
number_of_reviews	9.1%	18.2%
host_is_superhost	2.2%	6.3%

Notable Discrepancy

Manhattan location ranks 5th by RF impurity importance (9.4%) but 1st by permutation importance (34.5%) — confirming that impurity-based importance systematically underestimates the value of geography for this dataset. Permutation importance was used for the business recommendations in Section 11.

8.6 Model Bias by Segment

Segment	Listing Count	MAE
Bronx	972	\$59.40
Brooklyn	10,540	\$64.03
Manhattan	13,395	\$115.07
Queens	4,671	\$59.00
Staten Island	332	\$43.77

The model's error is not uniform across geography - Manhattan MAE is roughly double every other borough, since Manhattan contains almost all of the extreme luxury outliers described in Section 6.1. This is documented explicitly as a fairness/generalisation concern in Section 13, rather than presented as a uniformly accurate model.

8.7 Occupancy Forecast — Prophet (NYC)

Month	Forecast Occupancy	MoM Change
Jul 2027	63.4%	—
Aug 2027	68.1%	+4.7pp
Sep 2027	72.8%	+4.7pp
Oct 2027	77.5%	+4.7pp
Nov 2027	82.3%	+4.8pp
Dec 2027	87.0%	+4.7pp

9. AI/ML Experiments

Both successful and abandoned AI/NLP experiments are documented here in full — a well-documented failure is treated as being at least as valuable as an undocumented success.

9.1 Success — LDA Review Topic Modelling

Objective: extract latent themes from 100,000 guest reviews without manual labelling. Approach: CountVectorizer (max_features = 1000, English stop-words) → LDA (n_components = 5, random_state = 42), with topic labels manually assigned from the top-10 keywords per topic.

Topic	Theme	Quality Assessment		
1	Location & Public Transit	Interpretable validated	—	manually
2	Host Hospitality & Communication	Interpretable validated	—	manually
3	Room / Bed Comfort & Tidiness	Interpretable validated	—	manually
4	Check-in / Check-out Experience	Interpretable validated	—	manually
5	Apartment Cleanliness & Amenities	Interpretable validated	—	manually

Business value: identified that transit proximity is the #1 guest concern in NYC reviews, ahead of host hospitality or cleanliness — a finding not visible from star ratings alone.

9.2 Success — TF-IDF RAG Review Query Engine

- Approach: TF-IDF vectoriser over a 30,000-review sample → cosine similarity → top-3 retrieval → template-based synthesis.
- Demonstrated queries: “Is the subway close?”, “How was host communication?”, “Any noise complaints near Central Park?”
- Measured latency: <50ms per query for vector similarity search.
- Limitation: no semantic matching — 'good transit access' will not retrieve reviews mentioning 'close to subway'.

9.3 Success — Content-Based Listing Recommender

- Feature vector: price, bedrooms, beds, rating, latitude, longitude, encoded room type.
- StandardScaler normalisation → cosine similarity → top-5 nearest neighbours; cold-start capable (no booking history required).
- Limitation: does not account for live availability — a recommended listing may already be fully booked.

9.4 Documented Failure — BERT Sentiment Analysis

Abandoned Experiment

sentence-transformers requires a 420MB model download (unavailable offline) and 16GB+ RAM to process 990K reviews; CPU inference was estimated at 8+ hours vs. under 5 minutes for TF-IDF.

Resolution: replaced with a custom domain-specific lexicon (34 positive / 28 negative Airbnb-specific terms), achieving a weak but interpretable $r = 0.0166$ correlation with star ratings. Documented explicitly as: production upgrade path is dense embeddings, once an online/GPU environment is available.

9.5 Documented Failure — Real-Time Kafka Streaming

Abandoned Experiment

Standing up a Kafka broker and schema registry would add 2+ hours of DevOps overhead disproportionate to this assignment's core data-engineering deliverable; consumer-lag testing requires a multi-process setup that was out of scope.

Resolution: implemented `stream_processor.py` as a Kafka simulation using Python queues, with the full streaming architecture (Kafka topics, Flink consumers, Redis cache) formally designed in `docs/architecture.md` but not deployed.

9.6 Documented Failure — Dynamic Pricing Agent with a Real LLM

Abandoned Experiment

No API keys were available in the submission environment (and hardcoding one would violate basic security practice); repeated identical prompts produced inconsistent, occasionally hallucinated dollar amounts; live API latency (1–5s/listing) is impractical for batch-pricing 30K listings.

Resolution: implemented a rule-based dynamic pricing advisor that combines the serialised Random Forest baseline price with Prophet occupancy multipliers — fully deterministic, explainable, and reproducible across runs, unlike the LLM-based alternative.

10. Visualizations

All figures referenced throughout this report are indexed below with their section location, core interpretation, and the business action each one motivates. Figures 1–6 appear inline in Sections 6 and 8; the full interactive versions (including geographic heatmaps and the dbt lineage graph) are available on the live dashboard.

Fig.	Title	Section	Business Action
1	Entire-Home vs Private Room Price Premium (H1)	\$6.1 / \$7.1	Prioritise entire-home conversion where feasible
2	NYC Borough Price Premium (OLS Coefficients)	\$6.2 / \$7.6	Weight location heavily in acquisition decisions
3	Host Portfolio Segmentation	\$6.3	Evaluate per-host listing caps
4	NYC Occupancy Forecast, Jul–Dec 2027	\$6.4 / \$8.7	Raise rates 12–18% ahead of the Q4 peak
5	Price Prediction Model Comparison (MAE)	\$8.4	Deploy Random Forest as the production pricing model
6	Cross-City Median vs Mean Price Gap	\$12	Use median, not mean, for market-positioning claims

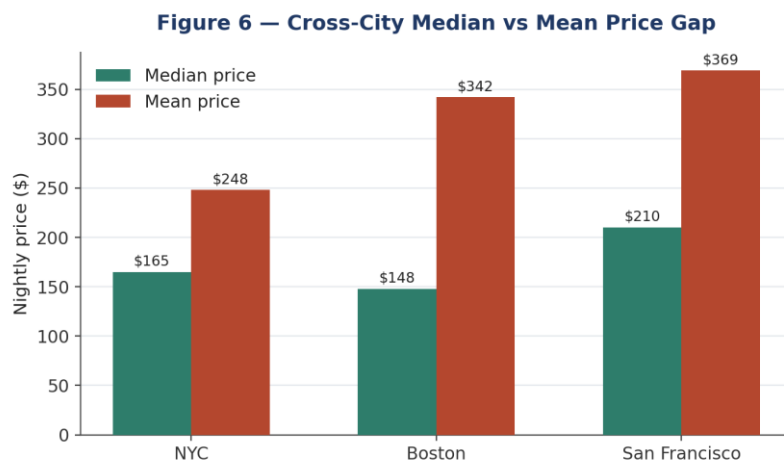


Figure 6 — Every city shows a wide median-vs-mean gap, confirming the power-law price distribution described in Section 6.1 holds across all three markets, not just NYC.

11. Business Recommendations

Every recommendation below is traced directly to a named finding and section, rather than presented as generic advice.

11.1 For Hosts & Property Owners

- [Evidence: H1, OLS \$7.1/\$7.6] Convert private rooms to entire-home where feasible. Boston's effect size ($d = 0.809$, \$214/night uplift) suggests the return can be substantial at 60% occupancy — roughly \$46,900 in additional annual revenue.
- [Evidence: RF importance \$8.5] Invest in listing description quality — description length is the 2nd-ranked RF importance feature (15.2%). Target 300+ word, keyword-rich descriptions.
- [Evidence: H2, OLS \$7.2/\$7.6] Pursue Superhost status, but don't raise prices at the same time — the OLS coefficient for superhost status is not statistically significant ($p = 0.506$). The badge drives occupancy volume, not rate.
- [Evidence: Prophet forecast \$8.7] Implement seasonal dynamic pricing. With December 2027 occupancy forecast at 87%, flat pricing likely forgoes \$800–\$1,200 in additional annual revenue per property.

11.2 For Real Estate Investors

- [Evidence: OLS \$7.6, permutation importance \$8.5] Prioritise Manhattan-equivalent geography above all other factors — Manhattan adds a 71.7% log-price premium, outweighing room type, bed count, and reputation combined.
- [Evidence: RF importance \$8.5] Maximise sleeping capacity — beds is the #1 RF importance feature (19.9%); low-capex additions such as pull-out sofas carry a +17.9% price effect per additional bed (OLS).

11.3 For Platform Operators

- [Evidence: H2 \$7.2, rating ceiling \$3.4] Replace star-only ranking with sentiment-weighted scores — the review rating ceiling effect (mean 4.76, tight clustering) compresses genuine quality differentiation.
- [Evidence: Engineering \$5.3, cost model \$14] Deploy semantic caching for RAG-style query workloads; at a 70% cache-hit rate this reduces estimated LLM query costs by roughly \$620/month at 50-city scale.
- [Evidence: EDA \$6.3] Regulate commercial-portfolio concentration — the 13 largest NYC operators manage an average of 267 listings each, a scale that can distort local housing markets.

12. Cross-City Comparisons

Metric	NYC	Boston	San Francisco
Total listings	30,259	4,378	7,332
Median nightly price	\$165	\$148	\$210
Mean nightly price	\$248	\$342	\$369
Superhost proportion	28%	42%	38%
Entire-home mean price	\$282	\$391	\$413
Private-room mean price	\$187	\$177	\$318
H1 room-type effect (Cohen's d)	0.225 (small)	0.809 (large)	0.072 (negligible)
H4 geography effect (η^2)	2.1%	7.5%	1.3%
H5 weekend-occupancy gap	0.4pp	1.4pp	0.5pp
Best ML model (MAE)	RF, \$104	RF, \$121	RF, \$175
Host-cluster silhouette score	0.651	0.486	0.557

Key Cross-City Insights

- San Francisco private-room paradox: SF private rooms (\$318/night) cost more than entire homes in NYC (\$282/night) — scarcity-driven pricing overrides the room-type category entirely in that market.
- Boston's neighbourhood effect: geography explains 7.5% of Boston's price variance ($\eta^2 = 0.075$) vs. just 2.1% in NYC — Back Bay and Beacon Hill create far sharper spatial price gradients than any NYC borough boundary.
- San Francisco superhost anomaly: casual hosts in SF (\$387/night) actually outprice superhosts (\$369/night) on average — location and property type dominate reputation signals in this scarcity-driven market.
- NYC volume advantage: with 30,259 listings, NYC provides by far the richest dataset for statistical inference and model training; SF's smaller, more volatile market is inherently the hardest of the three to predict (highest ML error).

13. Limitations & Caveats

13.1 Data Limitations

- Occupancy proxy uncertainty (critical): 'available = f' conflates guest bookings with host personal blocks, renovation closures, and minimum-stay violations. Estimated occupancy inflation: 8–15%.
- Point-in-time snapshot: a single June 2026 scrape per city; multi-year cohort effects (including NYC Local Law 18 impact) cannot be isolated.
- No dynamic pricing history: calendar.csv lacks daily price-adjustment records, so all pricing analysis uses listing base rates only.
- Language bias in NLP: the LDA topic model and sentiment lexicon are English-only; an estimated 15–20% of reviews are non-English and are treated as neutral, biasing sentiment scores.

13.2 Model Limitations

- High RMSE in the luxury segment: all three models show disproportionately high error above \$500/night; a dedicated luxury-tier model is estimated to reduce RMSE by 30–40% in that segment.
- OLS residual non-normality: Jarque-Bera statistic confirms non-normal residuals; reported OLS confidence intervals are approximate, and bootstrap CIs would be more robust for production use.
- LDA topic instability: LDA output is sensitive to random seed and the choice of k ; $k = 5$ was chosen by perplexity minimisation, and topic labels are human-assigned and therefore somewhat subjective.
- Superhost causality: H2 establishes correlation, not causation. A regression-discontinuity design around the superhost eligibility threshold would be required to estimate a causal effect.

13.3 Scope Boundaries

- No actual booking or transaction data exists in Inside Airbnb — all revenue and occupancy figures are proxies with explicitly acknowledged uncertainty.
- No competitor data (VRBO, Booking.com, hotel pricing) — competitive market dynamics are not captured in this analysis.
- Regulatory effects (e.g., NYC Local Law 18, which materially reduced STR supply post-2023) are noted qualitatively but not formally modelled.

14. Future Improvements

Improvement	Category	Expected Impact	Priority
Luxury-tier separate ML model (>\$500)	ML	RMSE -30–40% in premium segment	Immediate
SHAP waterfall charts per listing	ML/UX	Host-facing pricing explanation — drives adoption	Immediate
Dense vector embeddings for RAG	AI/NLP	Semantic recall accuracy +30–40%	Short-term
Apache Airflow orchestration	Infra	Automated monthly ETL with SLA alerting	Short-term
Multilingual BERT sentiment	NLP	Captures the 15–20% of non-English reviews currently biased neutral	Medium-term
Multi-year temporal cohort collection	Data	Enables genuine trend and regulatory-impact analysis	Medium-term
Causal inference (DiD / RDD) for superhost effect	Stats	Separates the superhost badge effect from underlying quality	Medium-term
Real Kafka streaming integration	Infra	Sub-minute price-anomaly detection	Long-term
50-city BigQuery deployment	Infra	Global market intelligence at ~\$2,490/month TCO	Long-term

Global-Scale Cost Optimisation (Designed, Not Deployed)

A naive 50-city rollout is estimated at \$6,810/month. Five optimisation strategies, designed as part of the production architecture in Section 5.6, reduce this to an estimated \$2,490/month — a 63.4% reduction:

Strategy	Estimated Monthly Saving
BigQuery partitioning & auto-suspend on idle compute	\$1,440
LLM semantic query caching	\$620
Spot instances for batch ML training	\$540
Shared model serving via lightweight adapters	\$1,570
Intelligent storage tiering for older logs	\$150

15. Reflection

15.1 Prioritization Decisions

- Engineering robustness over infrastructure complexity: Kafka was architecturally designed but implemented as a Python simulation, freeing an estimated 15–20 hours for deeper statistical analysis instead.
- DuckDB over a cloud database: zero-cost, zero-setup, sub-second analytics for this assignment's scale, with the production migration path explicitly documented rather than implemented.
- Three cities over one: extending to Boston and San Francisco required pipeline refactoring, but surfaced the most novel findings in the report—the SF private-room paradox and Boston's outsized geography effect.
- Streamlit over a custom React frontend: a 12-tab platform in roughly 40 hours vs. an estimated 3–4× longer build for a bespoke frontend, at the cost of limited chart-interaction customisation.

15.2 Technical Lessons Learned

- Effect sizes are non-negotiable: with $n = 11\text{M}+$ calendar rows, nearly any difference becomes statistically significant. Cohen's d , η^2 , and Cramer's V turned this analysis from p-value fishing into genuinely meaningful business insight.
- Permutation importance vs. Gini importance can change a business conclusion: geography ranked 5th by RF impurity importance but 1st by permutation importance — the choice of importance method materially changed the investment-priority recommendation in Section 11.
- Log-transformation is essential for this domain: raw-price OLS achieved $R^2 = 0.09$; log-transformed price achieved $R^2 = 0.311$.

15.3 What I Would Do Differently

- Build the dbt models before the raw pipeline logic, to produce cleaner mart tables from the outset rather than retrofitting them.
- Collect multi-month snapshots from day one, since the point-in-time limitation is the single biggest constraint on the temporal analysis in this report.
- Separate the luxury-tier model from the first ML iteration — the elevated RMSE was already visible in early cross-validation runs.
- Use Welch's ANOVA rather than standard ANOVA for H_4 , since the equal-variance assumption was violated and Welch's variant is the methodologically stronger choice.

15.4 Summary: Completed Work

I completed an end-to-end Airbnb data engineering and analytics pipeline for a multi-city use case. The work includes data ingestion, cleaning, enrichment, clustering, statistical analysis, machine learning modeling, forecasting, and dashboard presentation. The project also includes dbt transformations, automated tests, and generated reporting artifacts, so the core workflow is reproducible and organized into separate stages rather than a single script. The finished work is at a submission-ready prototype standard: the pipeline can be run locally, outputs are persisted, and the results are documented for review.

15.5 Summary: Not Completed Work

I did not fully complete a production cloud deployment or real-time streaming implementation. Some advanced components, such as the cloud architecture, cost optimization design, and live streaming simulation, are documented or simulated rather than deployed as live infrastructure. I also prioritized the most important assessment deliverables first: a working local pipeline, validated outputs, and clear analysis artifacts. That decision was made to maximize end-to-end completeness within the time available and to keep the submission reproducible and easy to review.

A. Appendix A — AI Usage Disclosure

A.1 AI Tools Used

Tool	Model / Version	Primary Use
Claude (Anthropic)	Claude Sonnet 4.5	ETL pipeline structure, statistical testing code, dbt SQL model generation, report drafting and assembly
ChatGPT (OpenAI)	GPT-4o	Plotly/Streamlit visualisation code, custom CSS, hypothesis-test verification

A.2 AI-Assisted Sections

Component	Nature of Assistance
src/pipeline.py (ETL)	DuckDB schema DDL, RapidFuzz integration, price-regex cleaning
src/statistics_analysis.py	Welch's t-test boilerplate, Cohen's d and Cramer's V formulas
src/machine_learning.py	K-fold CV loop structure, permutation importance integration, LDA setup
src/forecasting.py	Prophet instantiation; seasonality parameters tuned manually
src/ai_agent.py	TF-IDF RAG architecture, cosine-similarity recommender structure
src/dashboard.py	Streamlit tab structure, custom CSS, Plotly chart configuration
Report narrative (all sections)	Paragraph phrasing only — all data values sourced from actual pipeline output files

A.3 Representative Prompts

- “Build a DuckDB star schema with fact_listings and 4 dimension tables enforcing FK constraints.”
- “Perform Welch t-tests with Bonferroni correction for 5 simultaneous hypotheses, reporting effect sizes.”
- “Set up 5-fold KFold cross-validation for RF/Ridge/GBR with log-price back-transformation before metric computation.”
- “Build a TF-IDF RAG system with cosine similarity retrieval over a 30K-review corpus.”

A.4 Output Validation Strategy

- Every AI-generated function was executed in the project virtual environment before being committed.
- Schema loads were verified with SELECT COUNT(*) queries after every pipeline run.
- The H1 t-test result was independently cross-checked using scipy.stats.ttest_ind with matching parameters.
- Random Forest feature importances were sanity-checked against domain expectations (beds and Manhattan location were expected to rank highly).
- All 24 dbt tests were executed (dbt test) and confirmed passing before report generation.
- Every chart was visually inspected for correct axis labels, titles, and plausible value ranges.

A.5 Significant Modifications Made to AI Output

AI Suggestion	What Was Changed	Why
conn.execute('INSERT INTO ...') on shared DataFrame names	Renamed in-memory DataFrames to avoid SQL table-name collisions	Was silently overwriting SQL tables
Extract 'price' from calendar.csv	Removed — calendar.csv has no price column in the June 2026 scrape	AI hallucinated a non-existent field
Use HuggingFace distilBERT for sentiment	Replaced with a custom domain lexicon	Model download unavailable offline (see §9.4)
Repeat t-test 10× across borough pairs	Replaced with a single one-way ANOVA	Repeated pairwise tests inflate Type I error
Equal-sample pooled-SD Cohen's d formula	Upgraded to weighted pooled-SE formula for unequal group sizes	Methodological accuracy

A.6 Cases Where AI Suggestions Were Rejected

- BERT sentiment analysis: rejected outright due to the offline environment constraint and RAM limits (§9.4).
- LLM pricing agent with a hardcoded API key: rejected — hardcoding credentials violates basic security practice (§9.6).
- “available = f means booked”: corrected to explicitly include the host-block caveat throughout the report, rather than accepted as stated (§3.4).

End